

Technical Report: Adapting Pretrained Models for Object Recognition in Low-Light Conditions via Gradual Unfreezing

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1.0 Introduction: The Challenge of Nighttime Computer Vision

Reliable computer vision is strategically important across numerous applications, but its effectiveness is often compromised in low-light environments. Nighttime or poorly lit conditions introduce significant challenges for standard recognition models, including low contrast, increased sensor noise, and a loss of color information, which collectively degrade feature distinctiveness. To overcome the need for massive, specialized datasets and extensive computational resources, this report investigates the use of transfer learning as a practical and powerful solution.

Transfer learning leverages the knowledge a model gains from a large, general-purpose dataset (such as ImageNet) and applies it to a different but related, often smaller, dataset. Models trained on ImageNet learn a rich hierarchy of visual features. Early layers learn to detect universal, "low-level" features like edges, blobs, and color gradients. Deeper layers combine these to identify more complex, "high-level" features that are specific to objects, such as car wheels or dog ears. This pretrained foundation allows a model to bypass the difficult initial "learning to see" phase and move directly to the task-specific "learning to distinguish" phase, enabling the development of highly effective models without the prohibitive cost of training from scratch.

A central challenge in this process is managing the "domain shift" between the source and target datasets. In this experiment, the source domain consists of the millions of predominantly well-lit, daylight images in ImageNet, while the target domain is a dataset composed exclusively of nighttime images. The core problem, therefore, is to effectively adapt the model's learned features across this significant visual and statistical gap.

The purpose of this report is to document and analyze a three-stage, gradual unfreezing methodology applied to a ResNet-18 model. This structured approach is designed to fine-tune the model for a specialized nighttime object recognition task,

maximizing performance by carefully balancing knowledge retention with domain-specific adaptation. The following sections detail the experimental setup, the rationale behind each training stage, and a quantitative analysis of the results.

2.0 Experimental Methodology

The experiment was designed to fine-tune a standard ResNet-18 model, pretrained on the ImageNet dataset, for a 10-class object recognition task. The target dataset, `codan_test_night`, consists entirely of images captured in low-light conditions, presenting a distinct domain shift from the model's original training data. The methodology was designed to systematically isolate and adapt distinct layers of the network, balancing the preservation of general-purpose features with the need for domain-specific tuning.

The technical setup utilized the `torchvision.models.resnet18` architecture as the base model. The nighttime dataset was split into an 80% training set and a 20% validation set to monitor performance. The following subsections detail the execution and strategic rationale for each stage of the fine-tuning process.

2.1 Stage 1: Feature Extraction with a Frozen Backbone

The initial stage of training focuses on adapting the model to the new task without corrupting its powerful, pretrained feature hierarchy. To achieve this, the entire ResNet-18 convolutional backbone was frozen by setting its parameters to `requires_grad = False`. Only the newly added, randomly initialized final classification layer (`model.fc`) was trained.

The strategic rationale for this step is to protect the sophisticated weights learned from ImageNet. A new, untrained classification head generates large error gradients initially. If the entire network were trainable, these gradients could propagate backward and drastically alter the stable, low-level feature detectors. By freezing the backbone, the pretrained model acts as a fixed feature extractor. This approach allows the new classifier to safely learn how to interpret the existing, high-quality features for the 10 nighttime classes before any changes are made to the feature extractor itself.

- **Optimizer:** Adam
- **Learning Rate:** 1e-3

- **Epochs:** 5

2.2 Stage 2: High-Level Feature Adaptation

In the second stage, the training was expanded to include the final residual block of the ResNet-18 architecture (model.layer4), whose parameters were unfrozen alongside the classifier head. The earlier layers of the network remained frozen.

This gradual unfreezing strategy is based on the hierarchical nature of feature learning in convolutional neural networks. The earliest layers learn to detect universal, low-level features like edges and color gradients, which are valuable across most visual domains. The deeper layers, however, learn to combine these into high-level, semantic features that represent complex object parts. By unfreezing layer4 first, the model is able to adapt these more abstract feature detectors to the specific visual characteristics of nighttime objects while preserving the foundational, general-purpose detectors in the earlier layers.

- **Optimizer:** Adam
- **Learning Rate:** 1e-4
- **Epochs:** 3

2.3 Stage 3: Global Optimization via Full Fine-Tuning

The final stage involved unfreezing all layers of the model, making every parameter trainable (requires_grad = True). This allows for end-to-end, global optimization of the entire network.

The rationale for this stage is to permit small, holistic adjustments across the entire feature hierarchy. With the high-level layers already adapted and the classifier stabilized, the model can now safely refine even its earliest feature detectors to better suit the nuances of the nighttime dataset. A critical component of this stage is the use of a very low learning rate. This adheres to the established best practice for fine-tuning, where learning rate is decreased as more of the pretrained network becomes trainable to avoid destabilizing well-calibrated weights. This precaution is essential to mitigate the dual risks of Catastrophic Forgetting, where the model overwrites the robust general features learned from ImageNet, and overfitting to the smaller target dataset.

- **Optimizer:** Adam
- **Learning Rate:** 1e-5
- **Epochs:** 3

This structured, three-stage methodology provides a controlled process for adapting the model from a general to a specific domain, leading to the performance outcomes detailed in the following section.

3.0 Results and Performance Analysis

This section presents the quantitative results achieved at the conclusion of each training stage. The performance metrics offer a clear picture of the model's adaptation to the nighttime object recognition task and validate the effectiveness of the gradual unfreezing strategy in navigating the daylight-to-nighttime domain shift.

The table below summarizes the final validation metrics—Accuracy, Precision, Recall, and F1 Score—recorded after each of the three fine-tuning stages.

Table: Model Performance Across Training Stages

Stage	Accuracy	Precision	Recall	F1 Score
Stage 1: Head Only	0.7640	0.7733	0.7649	0.7610
Stage 2: Layer 4 + Head	0.8420	0.8439	0.8437	0.8425
Stage 3: Full Fine-Tune	0.8560	0.8549	0.8564	0.8554

The results demonstrate a clear and positive performance progression. Stage 1 established a strong baseline accuracy of 76.4%, confirming that the generic ImageNet features are highly relevant even in a challenging, out-of-domain context. Stage 2 yielded the most significant performance gain—an increase of 7.8 percentage points to 84.2% accuracy. Finally, Stage 3 provided a smaller, incremental improvement of 1.4 percentage points, reaching a final accuracy of 85.6%. This quantitatively demonstrates a pattern of diminishing returns as the model performed minor, global refinements.

These outcomes validate the gradual unfreezing strategy for bridging the significant domain shift between ImageNet and the low-light target data. Because nighttime

images alter high-level semantic features (e.g., object shapes in shadow, partial illumination) more than they alter fundamental low-level features (e.g., edges, textures), the adaptation of deeper layers logically yields the largest performance increase. The substantial jump in Stage 2 highlights that while the foundational features from ImageNet are invaluable, adapting the model's high-level "vision" is critical for correctly interpreting the unique pixel distributions and noise patterns inherent to the nighttime domain.

4.0 Recommendations for Performance Enhancement

While the implemented transfer learning methodology proved successful, further performance gains are likely achievable by addressing the core data challenges of nighttime imagery *before* they reach the model. By integrating targeted data pre-processing and augmentation techniques, the input data can be enhanced to make features more salient and the model more robust to variations in low-light conditions. The following strategies are recommended for future work:

- **Histogram Equalization (e.g., CLAHE):** This technique redistributes pixel intensities to increase global contrast, making under-exposed features more salient for the convolutional layers. For nighttime images, it can make features in dark or washed-out regions more distinct.
- **Gamma Correction:** This is a non-linear operation used to adjust image brightness. It can be applied to systematically brighten darker images or darken overexposed ones, helping to normalize illumination levels across the dataset.
- **Targeted Data Augmentation:** This improves model generalization by forcing the model to learn features that are invariant to common low-light artifacts like sensor noise and illumination shifts. Augmenting with Gaussian noise simulates camera sensor noise, while brightness jitter exposes the model to a wider range of lighting conditions.
- **Synthetic Data Generation (e.g., CycleGAN):** A more advanced approach involves using Generative Adversarial Networks for image-to-image translation. A model like CycleGAN could be trained to convert widely available daytime images into realistic nighttime equivalents, synthetically and significantly expanding the training dataset.

Implementing these data-centric enhancements could further improve the model's ability to generalize and perform accurately on challenging low-light data.

5.0 Conclusion

This report details a successful application of a three-stage gradual unfreezing methodology to adapt a pretrained ResNet-18 model for object recognition in low-light conditions. The experiment demonstrates that this structured approach is highly effective for bridging the significant domain shift between standard daylight imagery and a specialized nighttime dataset, ultimately achieving a final validation accuracy of 85.6%.

The core analytical insight from this work is that while transfer learning provides a powerful foundation, targeted fine-tuning is critical for achieving high performance on visually distinct domains. The largest performance gains were realized not by training the classifier alone, but by carefully unfreezing and adapting the model's high-level feature layers, allowing it to learn the specific characteristics of the target domain while retaining its general visual knowledge.

The experiment affirms the value of gradual unfreezing as a robust and resource-efficient technique for specialized computer vision tasks. Performance can likely be enhanced further by integrating the recommended data pre-processing and augmentation strategies to directly address the inherent challenges of low-light imagery.

6.0 Conclusion

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